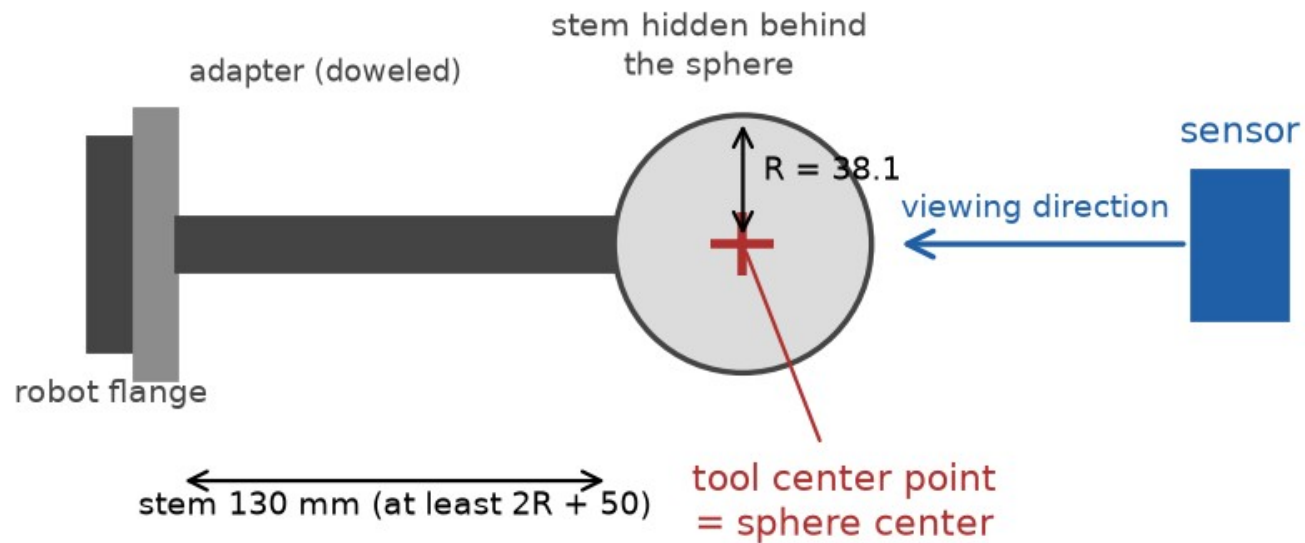


Stage-1 Calibration: Procurement and Build

The spheres, board and fixtures needed for the depth-sensor calibration captures



Two spheres and a board carry the known geometry



Sphere on its stem, side view: the stem points away from the sensor so that the sphere hides it.



Sphere A

76.2 mm diameter; used from 300 to 550 mm



Sphere B

152.4 mm diameter; used from 500 to 1,100 mm



Board

200 x 150 mm, at least 6 mm thick, flat to 0.05 mm after painting; tilted 0, 20 and 40 degrees

New equipment costs about \$1,900 to \$5,400

Item	Estimated cost (USD)
Sphere A, bearing ball, bonded stem, blasted, measured	\$350 to \$850
Sphere B, turned aluminum, anodized, measured	\$600 to \$1,800
Stems, two	\$150 to \$400
Adapter plates, two	\$250 to \$600
Board, glass or ground aluminum, painted	\$50 to \$400
Board adapter	\$250 to \$600
Three-ball nest	\$150 to \$450
Dial indicator with magnetic base	\$100 to \$300

\$1,900 to \$5,400

bearing-ball sphere A, aluminum sphere B

\$3,550 to \$10,250

both spheres made to order

Estimates of October 2026, plus or minus 50 percent; sphere B is the least certain. The sensor, robot and capture computer are in hand.

Spheres must be matte and measured, not certified



Finish

- Every bearing ball arrives mirror-bright: unusable
- Bead-blast with fine glass bead: removes micrometers
- Matte paint is second choice: 0.02 to 0.05 mm per coat
- "Chrome steel" is the 52100 alloy, not plating



Diameter

- Known to 0.025 mm, measured after the finish
- Micrometer at several orientations, or a CMM
- Aluminum grows 0.0035 mm per degree C: measure at session temperature
- The measured value goes in the pose log



Mounting

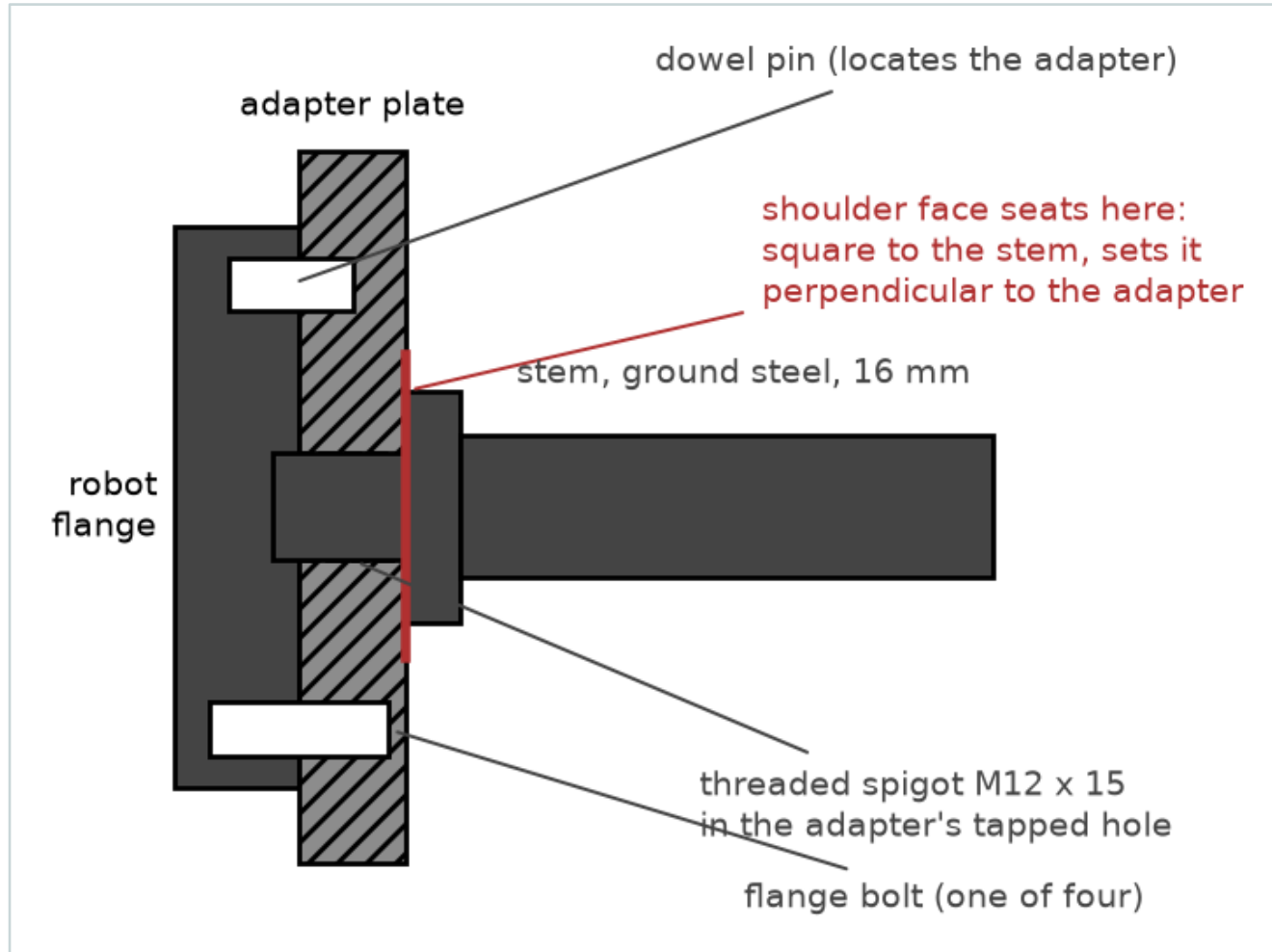
- Hardened balls cannot be tapped
- Steel ball: drilled or EDM blind hole, bonded stem
- Aluminum sphere B: thread cut when it is turned
- No pre-tapped precision sphere exists above 1.5 inch

Candidate suppliers

Item	Candidate suppliers	Notes
Spheres, made to order	Bal-tec; RGP Balls; Industrial Tectonics; Morgan Advanced Materials	3 and 6 inch matte spheres with a mounting hole are custom items
Matte reference spheres, stock	MetrologyWorks; Rothbucher RSLB10M (Baseline Equipment)	MetrologyWorks up to 1.5 inch; Rothbucher 145 mm, sphericity 0.3 mm
Board plate	McMaster-Carr MIC-6 plus a grinding shop; Edmund Optics float glass	Painted matte after grinding; flatness checked after mounting
Nest balls	McMaster-Carr; Bal-tec	Grade-25 bearing balls
Adapters, stems, sphere holes	Local machine shop; EDM shop for hardened balls	Hardened balls cannot be tapped
Diameter measurement	Calibration laboratory with a coordinate measuring machine	Measure after the matte finish

From suppliers' web pages, October 2026; not an endorsement. Confirm size, finish, measurement and thread before ordering.

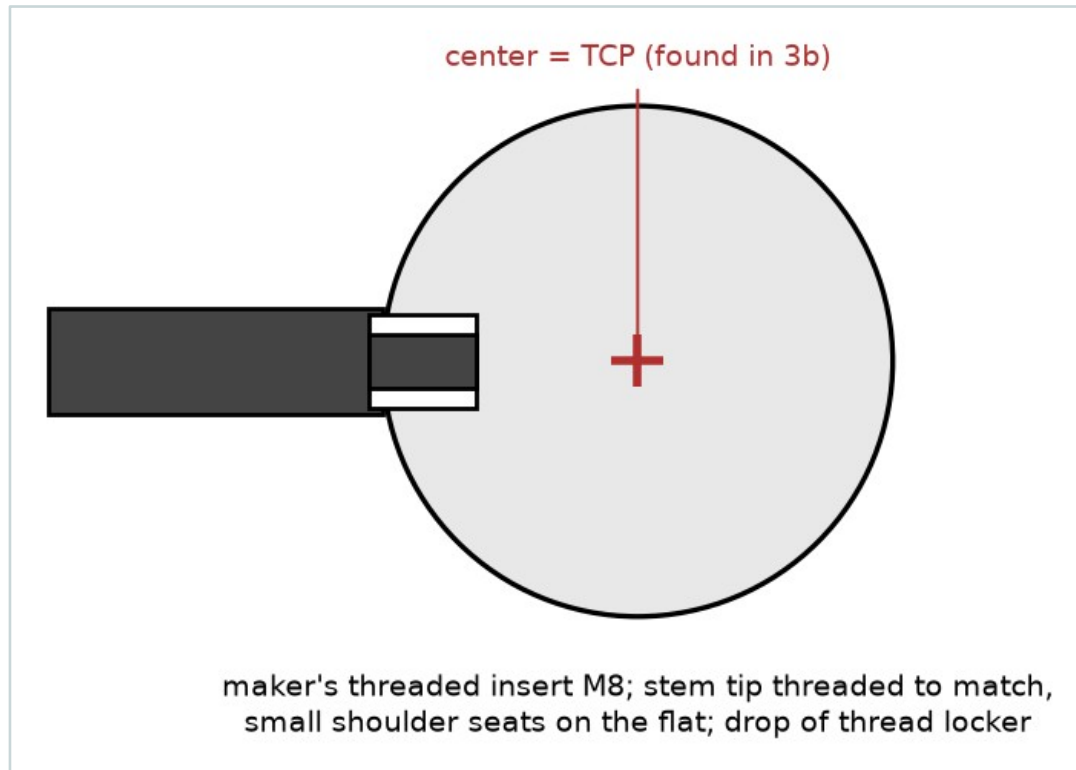
A stiff, repeatable chain from flange to sphere



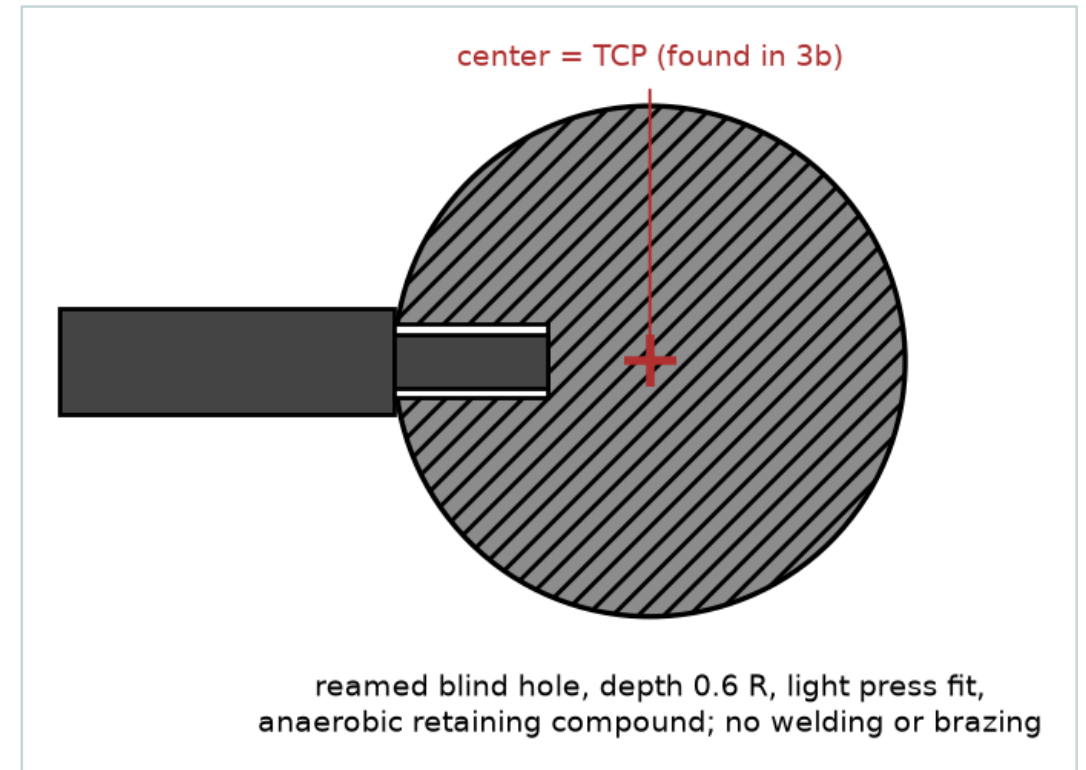
- Adapter located by the flange dowel, so a remount lands in the same place
- Ground stem, 16 mm for A and 30 mm for B, keeps sag to 0.02 to 0.05 mm
- Stem length at least $2R + 50$ mm, so the sphere hides the stem
- Witness mark across stem and adapter shows a loosened joint
- After any remount, re-check in the nest: readings within 0.1 mm

Adapter on the flange's dowel, with the stem's shoulder seated on the adapter face.

The sphere end: threaded insert or bonded stem



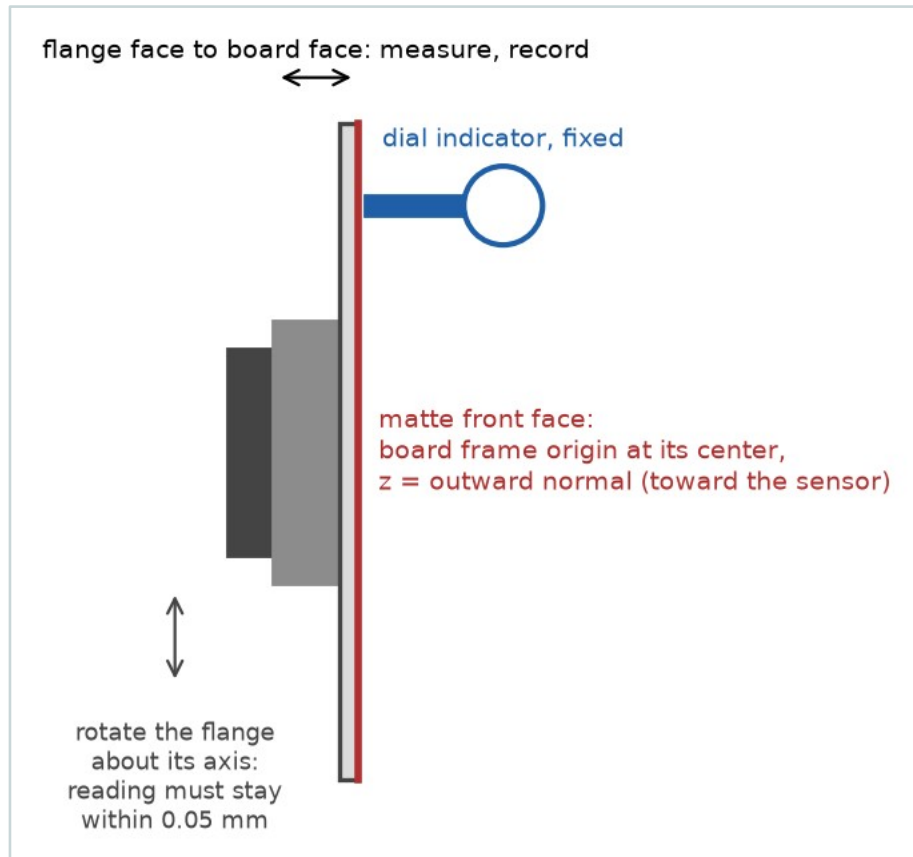
Ceramic sphere: maker's threaded insert



Steel sphere: bonded blind hole

The hole need not be exactly radial: the tool-center-point routine measures the center wherever it ends up.

The board must stay flat after finishing and mounting



Board on its adapter. The runout and frame checks shown here are done at setup (deck 2).



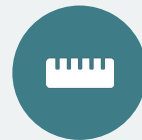
Plate

200 x 150 mm, at least 6 mm thick; ground aluminum tooling plate or float glass



Finish

Matte light gray paint on the front face; do not bead-blast a plate thinner than about 10 mm, it bows



Flatness

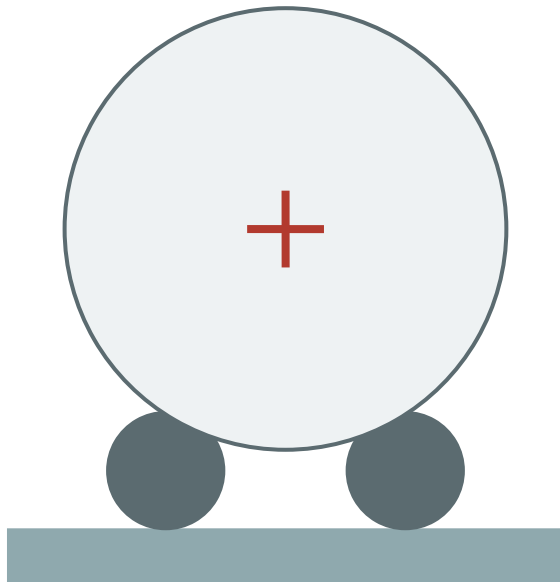
0.05 mm, checked after painting and mounting, not before; keep the flatness report



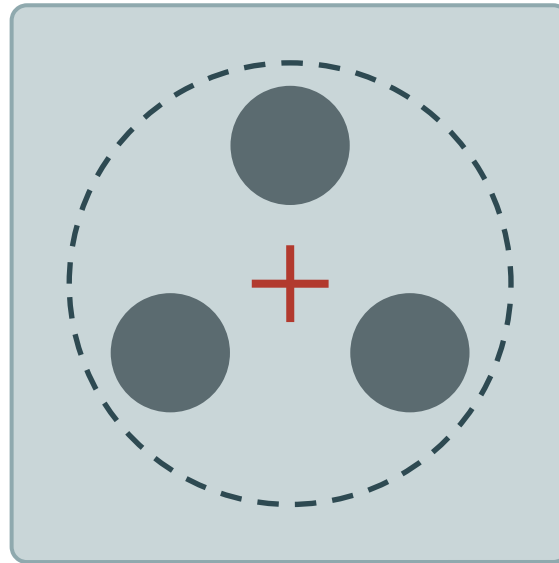
Adapter

Two dowels and a clamp hold the board square to the flange axis, centered on it, and in the same place after a remount

A three-ball nest holds each sphere at a fixed center



Side view



Top view



Three hardened balls, about 24 mm diameter, pressed into a base



Ordinary grade-25 bearing balls are good enough: the nest's quality comes from the balls being rigidly fixed



Bolted to the table where the robot reaches with the wrist down and tilted about 40 degrees to either side



Used once per sphere at setup, and after any remount of an adapter or stem

Checks before the fixtures go to the robot cell

- ✓ Adapter tapped hole square to its mounting face within 0.02 mm over 100 mm
- ✓ Each stem straight: roll it on a flat surface
- ✓ Both spheres matte, uniform and free of highlights
- ✓ Each sphere diameter measured after finishing, with the temperature recorded
- ✓ Board flat to 0.05 mm after painting and mounting
- ✓ Witness marks across each stem-adapter joint
- ✓ Robot payload rating covers sphere, stem and adapter together

